

Rehabilitation in sport medicine

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Introduction

Physical activity is associated with improved quality of life. However, being physically active carries the risk of injury and re-injury. At the present time, the incidence and recurrence of overuse injuries are high. Recent longitudinal studies in professional soccer players have shown that muscle injury rates and their recurrence rates increased during 11 consecutive seasons by 4% annually from 2001 to 2014, despite ongoing scientific investigation as well as improved educational background and knowledge of the different professionals involved in the rehabilitation process (Ekstrand et al., 2013). Therefore, there is a substantial interest in sports medicine to improve the prevention, treatment and rehabilitation of sports-related injuries.

The specific training of fascial tissues is suggested to lead the augmentation of the mechanical efficiency of the musculoskeletal system. However, the rise of external loads required to increase sport performance does also represent an additional stress factor in the medium term, which may eventually be associated with an increase of the risk of sport-related injury due to mechanical overuse. Even in the short term, tissue repair processes following a traumatic acute injury modify the mechanical properties of soft tissue around the scar area in a more direct and precise way. As a consequence, the involvement of fascial tissue in human movement, as regards its capacity to

absorb and transmit mechanical forces via different musculoskeletal structures, may be distorted and, hence, be the origin in the development of local and remote dysfunctions. From a therapeutic standpoint, skillful application of manual forces to the fascial system may help to condition and reverse collagen overproduction, thus improving tissue functionality and optimizing rehabilitation of musculoskeletal injuries.

Against this background, the boom in scientific knowledge on myofascial tissues and its potential clinical implications in the physiopathology of sport-related injuries invites us to continue exploring and developing new ways to detect, treat and control the mechanical properties of connective tissue.

Tissue adaptation following overuse

Mechanotransduction is the means by which external mechanical loads across the fascial network (shearing, tension and/or torsion) are transduced into biochemical responses at an effector cell level, generating structural and functional modifications on different musculoskeletal system components (Kjaer, 2004). Optimization of sport performance necessarily entails an increase of training loading and a decrease of rest periods, which drive a process of physiological adaptation associated with progressive and steady changes in the muscle, the muscle fascia and the tendon (Kjaer, 2004; Miller

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et al., 2005). Nonetheless, different papers suggest that the mechano-adaptation of the connective tissue after frequent high-intensity loadings includes a persistent and low-grade inflammatory reaction around the extracellular matrix, even in the absence of structural alterations on the macro-level (Barbe et al., 2013; Zügel et al., 2018). Such micro-inflammation is linked to an extended release of several proinflammatory and nociceptive substances, which, in the case of some cytokines (interleukin 1 β tumour necrosis factor and, mainly, TGF β 1), cause tissue stiffness through distinction and proliferation of fibroblasts as well as excessive collagen synthesis (Barbe et al., 2013; Zügel et al., 2018).

Mechanical tissue behavior following overuse

Structural changes in muscle fascia micro-architecture, and especially in the configuration of joints and intramuscular septa connecting adjacent muscles, are crucial in force distribution since they increase pre-tension in the deep fascia and optimize longitudinal force transmission to the tendon (Huijing, 2009; Bernabei et al., 2017). When no discomfort or pain appears, and when it is associated with high sports performance, the decrease of viscoelasticity and force absorption capacity must be considered as a physiological adaptation helping to optimize force transmission between muscles and tendons. However, in the medium term, the ensuing restructuring process could give rise to a stiffer fascial network, hence decisively modifying the soft tissue's viscoelastic behavior. In this context, decreases of local elasticity limit the fascial system's capacity for deformation and force absorption, which, in turn, magnifies the risk of musculoskeletal injury (Purslow, 2010). In sum, continuous high-intensity loading, reduced rest periods and an inadequate monitoring of the

mechanical connective tissue properties, may result in a lower tolerance threshold to musculoskeletal system loading, particularly during eccentric movements. Therefore, it is necessary to understand the association between tissues of the fascial system and the origin of muscular, tendinous and joint injuries related to sport practice.

Relevance of fascial stiffness in selected overuse injuries

Myofascial (muscle) injury

Different modifiable factors in the prevention and treatment of muscle injury have been suggested, including, but not limited to, muscle fatigue, muscle strength imbalance, neuromuscular inhibition, poor motor control of the pelvic and trunk muscles, limited flexibility or premature return to play (Silder et al., 2010). In addition, considering that the susceptibility for sustaining a muscle strain injury is greater in high-force eccentric contractions from a lengthened position, adequate viscoelastic behavior of soft tissues would be essential to dissipate and absorb the kinetic energy required during eccentric terminal movements, as well as during external loading via foot-ground contact (Chumanov et al., 2011) (Chapter 16).

Tendon injury

The increase of fascial stiffness may directly influence the development of tendinous disorders, since the tendon is directly connected to the deep fascia (Dalmau-Pastor et al., 2014; Grob et al., 2016). Indeed, new research shows tendon architecture as a specialized structure made of terminal layers of connective tissue arranged in the shape of a spiral with disruption between a profound muscle fascia (paratendon) and intra- and intermuscular connective tissue, as well as intermuscular septa

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and other collagen-reinforced structures. The connective tissue provides important stiff connections between agonist and antagonist muscles, from which force is transmitted to the tendon (Grob et al., 2016). Thus, the force generated by the muscle that is transferred to bones via tendons to produce movement is not only transmitted by each myofibril, made up of sarcomeres arranged in series and equipped with a myotendinous junction (myotendinous force transmission), but also by the existing fascial continuity between muscle fascia and tendon. This important functional continuity hence enables force transmission via the muscular supporting connective tissue (myofascial force transmission) (Huijing et al., 2009). As stated above, mechanical loading at physiological levels is beneficial to tendons, which directly respond to physical activity by increased metabolic activity and collagen synthesis. However, the mechanical properties of tendons, including their Young's modulus (a measure of the stiffness of an elastic material), may undergo modifications in response to excessive loading, modifying the adequate stress-strain balance (Järvinen et al., 2005 A).

Joint injury

Contraction and activation of the different muscle groups lead (1) to an increase of joint compressive forces through muscle shortening, and (2) to the augmentation of stiffness in peri-articular tissues (Barker et al., 2014). Also, higher pre-tension of the fascial system and reduced deformability of peripheral structures, transmitting mechanical tension to the articulation, could be related to an increase of the physiological compression forces at joint level as well as to a reduction in the quality of accessory movements (gliding, bearing, translation and/or rotation). Indeed, an article showed a higher level of stiffness and thickness of external peripatellar structures in patients with patellar chondropathy

when compared with asymptomatic controls. This finding has been related to the increase of pain in joints due to stronger compression forces in patellofemoral joint level (Schoots et al., 2013).

Relevance of fascial stiffness in sports injuries related to muscle healing

Once injured, muscle tissue undergoes a slow and spontaneous healing process that results in the formation of connective (scar) tissue, which is usually fibrotic and stiff. Thus, during the initial phases of the healing process, due to structural damage and/or surgery (iatrogenesis) within the area of ruptured and necrotized fibers, there is a proliferation of inflammatory cells stimulating fibroblast activity and collagen production around the scar (Järvinen et al., 2005 B). Initially, the scar tissue is characterized by a disorganized and random arrangement of the new collagen fibers. The absence of parallel fiber orientation relative to their original force transmission axis is a feature common to scars missing mechanical stimuli. In the long term, it can frequently be observed that presumed healed sport injuries are, in fact, highly scarred, increasing the overall mechanical stiffness of the myofibrous tissue it replaces (Järvinen et al., 2005). In addition, proliferation of scar tissue leads to greater strain in the affected structures during eccentric contractions, which could contribute to discomfort and pain, as well an increased re-injury risk during high speed and high amplitude movements (Silder et al., 2010).

Relevance of fascial stiffness in sensory motor alterations

Histological studies have demonstrated the presence of free nerve endings of A- δ and C fibers in fascia, which are polymodal receptors acting

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as unspecific mechanoreceptors and potential nociceptors when facing pressure and/or excessive mechanical tension (Taguchi et al., 2013; Schilder et al., 2014). Interestingly, the density of such receptors is three times higher in the lumbar fascia than in the deep spinal muscles (Barry et al., 2015). In addition to the muscular fascia, free nerve endings are also located in the continuous network of connective tissue found in joints (joint capsule, ligaments and subchondral bone), tendons (paratendon), bones (periosteum) and the nervous system (epineurium). Their concentration is higher than the number of specialized mechanoreceptors (Cabuk and Cabuk, 2016).

Significantly, various experimental studies prove the existence of a functional connection between changes in mechanical properties of the connective tissue and the nociceptive processes resulting from repetitive strains. The release of proinflammatory and pronociceptive substances not only leads to an increase of fibroblastic activity and tissue stiffness, but is also related to different stages of sensorimotor alterations, such as peripheral sensitization and force decreases (Barbe et al., 2013; Zugel et al., 2018).

Essentially, the mechanical forces generated by the musculoskeletal system are decisive for tissue remodeling. Thus, a correct control of loadings will improve mechanical efficiency and the tolerance threshold for mechanical stress. However, pathological adaptations related to stiffening or scarring may result in changes in joint range of motion and muscle-tendon force transmission, as well as an activation of nociceptive mechanisms. At this point, the following question arises: Could a stage of excessive fascial pre-tension (fascial stiffness) be modified by a focused manual treatment? And, from a global perspective, how could this treatment be integrated into classical models for prevention,

treatment and rehabilitation of musculoskeletal injuries?

Optimization of the rehabilitation process from a fascia-oriented perspective

It seems apparent that there is a need to apply skillful manual forces over the more rigid areas to restore and improve the fascial system's capacity to absorb and dissipate repetitive mechanical loads. At the same time and following a structural injury, it will be necessary to control the effects of arthrofibrosis, a well-known sequela of structural injuries after sports trauma, when treatment choice has involved immobilization and protection of the injured area for primary treatment or post-operative management (Figure 44.1).

Rehabilitation in myofascial (muscle) injury

The mechanisms of mechanical loading and mechanotransduction are well recognized in the treatment of muscle injuries as they can accelerate the recovery process due to activation/production of satellite cells and growth factors (Järvinen et al., 2005 B). Nonetheless, the healing process may be conditioned by an inadequate regeneration-repair equilibrium. Considering the high mechanosensitivity of fibroblasts, the early use of external loads may also cause an increase in collagen synthesis in the mechanical area of the scar as well as in adjacent associated tissues, leading to incomplete or unfunctional healing with decreased extensibility and eccentric force (Opar et al., 2012). Indeed, some specialists have proved a decrease of eccentric hamstring muscle force in athletes with reasonably high functional

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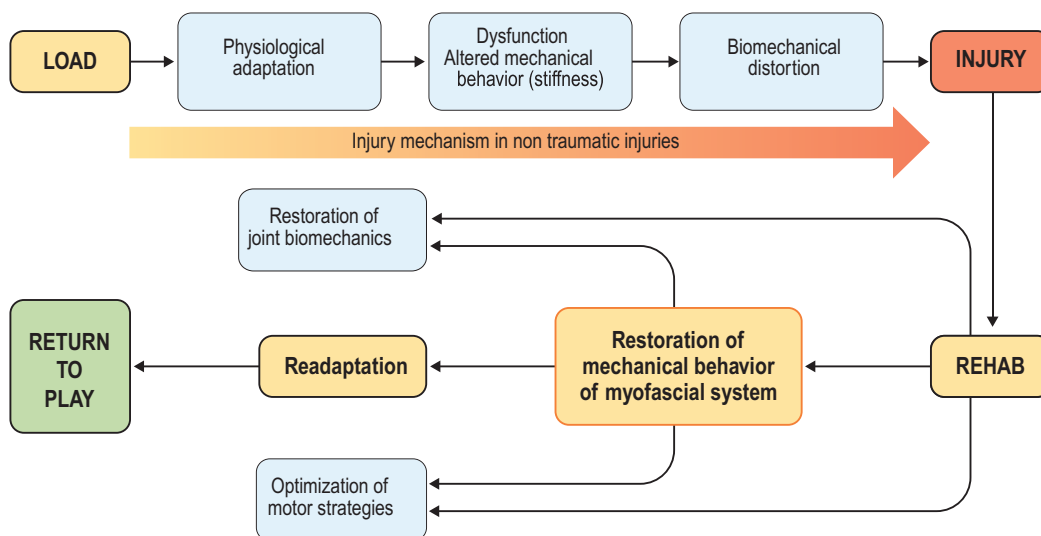


Figure 44.1 Injury mechanism in overuse injuries. Relevance of mechanical properties of fascial system during sports rehabilitation (rehab) physical and manual therapy

Mechanical loading and decrease of rest intervals renders the musculoskeletal system more load-resistant but at the same time it also considerably decreases its stress-susceptibility. In the medium term it can be related to biomechanical distortion and, consecutively, an increase in the risk of structural injury. We propose the need to optimize the process of sports rehabilitation, not only through the use of exercise-based programs, but also through the use of manual matrix remodeling techniques to optimize the mechanical behavior of the fascial system and the movement efficiency.

level (Sole et al., 2011). That study suggested the relevance, not only of neurophysiological factors (neuromuscular inhibition associated with pain or fear of recurrence), but also of mechanical factors of myofascial tissue in high eccentric movements. For this reason, beyond the fear-avoidance model and in line with the content of this chapter, we suggest the need to focus attention on both structural changes observed on echography or MRI and, most importantly, on the optimization of viscoelastic properties of scars, which would be essential to dissipate and absorb the kinetic energy needed for deceleration and high amplitude and high velocity movements (Martínez-Rodríguez and Galán del Río, 2013). This is why the optimization of tensile capacity of scars using the manual techniques described in this chapter may be a

way to lessen the excessive fascial stiffness of intermuscular septa and, more locally, the area around the scar.

Rehabilitation in tendon injury

With regard to the treatment of chronic tendon disorders, a variety of studies show the benefits of eccentric training programs (Murtaugh and Ihm, 2013). These include an increased loading tolerance, a higher rate of force development associated with a decrease in the traction supported by the tendon in the terminal eccentric movements, as well as an improved ability to store elastic energy. Even so, understanding the tendon as a collagenous terminal structure linked to the intramuscular connective tissue, and knowing the role of said connective tissue in the transmission of forces towards the tendon

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(Huijing, 2009), it must be remembered that tendon deformation not only depends on its own internal characteristics but also on the mechanical behavior of the deep fascia and intermuscular septa (Bernabei et al., 2017). Concurrently, the fascial system has been equipped with the capacity to dissipate and distribute force through the continuity of the fascial network. Thus, repetitive overuse initially causes cell matrix changes that thicken and harden the tendon (reactive tendinopathy). This becomes especially obvious in actions that combine high articular amplitude movements and eccentric contractions, during which a stiff fascial network is continually tensed by external forces (distancing of bone levers) and internal forces (muscular contraction). In the medium term, once micro-injured, the tendon undergoes a slow and spontaneous healing process (tendon disrepair) that results in the formation of scar tissue and a loss of normal collagen fiber organization. In this way, from the authors' clinical experience, prior to the execution of the eccentric and/or stretching exercises' protocols, the introduction of different manual techniques, as described below, is proposed to normalize the fascial network's tensional homeostasis and reduce the previously mentioned interfaces' stiffness.

Rehabilitation of bone and joint injuries

Historically, conventional rehabilitation programs base their treatment of decreased range of articular motion after an arthrofibrosis on the application of stretching as well as passive and active articular glide mobilization. However, within this context, the connective tissue intimately related to the joint (capsule, ligaments, periarticular myofascia) dehydrates, loses its elastic capacity, and presents considerable

disorganization (intercrossing collagen fibers and capsule shrinkage). As a result, deformation and sliding capacity between fascial layers are limited and, consequently, the movement capacity between articular surfaces is also reduced (gliding, bearing, translation and/or rotation). Recent research introduces the limitations of static stretching techniques for the optimization of the range of motion and the capacity of displacement within layers in lumbar fascia when restriction and fibrosis are detected (Langevin et al., 2018). Early and progressive exercise programs, together with structural techniques specifically designed to treat stiffer areas, are strongly recommended in this case. This said, and considering the anatomy of periarticular connective tissue and the direct relation of mechanical behavior and joint pain, a previous introduction to fascial tissue manipulation directly addressed to the fascial periarticular system should be proposed. This therapy would comprise different alternatives, such as matrix remodeling techniques, deep massage, deep friction and neuromuscular techniques, as well as several options of instrumental proceedings on restrictive and fibrotic areas. Such methodology, on the one hand, intends to induce the rehydration of the ground substance, enabling collagen fibers to move friction-free against each other, and, on the other, encourages the rupture of pathological collagen cross-links (Bove et al., 2016). Subsequently, the execution of progressive active loading exercises is suggested to improve the organization and arrangement of collagen fibers parallel to the main axes of mechanical tension.

It is important to highlight that stiffness within the periarticular myofascial tissue may alter muscle tone regulation and negatively influence protocols designed for muscle

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strengthening, proprioceptive re-education and training for restoration of sport-based movements. All three are required during the treatment process involving these types of injuries. This is intimately related to the role of the fascial tissue as a substrate of proprioception (mechanosensitive signaling system) and to the function of mechanoreceptors (Stecco et al., 2007; Van der Wal, 2009), specifically of muscle spindles, which are specialized structures of connective tissue within the muscle (endomysium and perimysium). They are characterized as being highly sensitive to even small tension variations. Therefore, considering that the main stimulus for these receptors is deformation, it can be inferred that an increase in stiffness and disorganization within the fascial system after immobilization protocols may decisively alter the capacity of adaptation in such a system. As a conclusion, the authors emphasize the importance of applying manual structural techniques so as to model the periarticular fascial tissue, to normalize the stimulatory mechanism of mechanoreceptors, thereby enabling effective efferent motor responses during muscle strengthening and proprioceptive protocols.

Manual matrix remodeling and control of fascial stiffness

To restore the tissue's normal viscoelastic behavior and allow the release of the elastic energy accumulated (not dissipated) within the surrounding area, thereby stimulating the extracellular matrix muscle to return from a high-tension to a low-tension state, the authors suggest the implementation of manual matrix remodeling techniques. These are based on the use of various mechanical stimuli to apply force through skin and superficial fascia until a deep

steady or progressive deformation of fascial tissue occurs by mixing torsion, shearing, traction, axial and/or compressive manual vectors. However, some fascial structures are of such thickness, fiber composition and resistance (iliotibial band, plantar fascia and other tendons of lower limb) that an immediate deformation may not be possible (Chaudhry et al., 2014). Due to this, and to modify the dynamic behavior of fascial tissue, we propose an approach over all fascia layers, connections, connective tissues, retinacula, septa and/or intramuscular trabecular systems. Although these have been traditionally disregarded by descriptive anatomy given their suggested low structural relevance, they directly intervene in mechanical tension transmission and force balance (Figure 44.2) (Huijing, 2009; Cruz-Montecinos et al., 2016; Bernabei et al., 2017). Principles of application in the use of matrix remodeling techniques can also be extended to stiff scar caused by excessive healing after a traumatic accident and/or surgical history (Martínez-Rodríguez and Galán del Río, 2013).

The following is a detailed description of the different phases of remodeling techniques for application in septa and scars (Figure 44.3):

- **Contact phase:** an initial compressive vector is provided with the second, third and fourth finger. Pressure must be applied progressively and should be enough to achieve a primary withstand level.
- **Stimulation phase:** at this stage, an axial, spiral and/or circular input is added to initial vector compression. The physiotherapist's fingers proceed to a deformation of tissue, slowly and continuously creating tension until a tensile barrier more resistant than the first is reached. The time that

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Figure 44.2 Image corresponding to the dissection of the crural fascia at the level of the vastus medialis

There is no solution of real continuity between fascia and muscle fibers, but there is continuity of fascial tissue to the inside of the muscle cells themselves. The separation muscle fascia is created by the analytical mind of the anatomist, determining in this way the understanding of the lesion mechanism, and consequently the manner in which we design the rehabilitation programs.

Image owned by Martínez y Galán, Faculty of Medicine and Health Sciences of Barcelona.

compression and torsion are applied simultaneously may vary (30–90 s).

- **Release phase:** in this phase, a release of elastic energy in the shape of “local unwind” may appear, depending on the response of the tissue after the initial stimulus. This is followed by a progressive decrease of original tension while there is still contact and reorganization of tissues in the restriction area, which leads to a natural repositioning of fingers as the barrier changes.

This process must be repeated as many times as necessary (usually from three to five times) until, eventually, an ease of “first tension feeling” is detected.

Re-adaptation and return to play

The processes of re-adaptation and retraining before return to play are usually based on progressive loading exercise programs adapted to the individual requirement of each athlete (core training, eccentric exercise and/or dynamic stabilization/muscle activation) and management of pain-avoidance behavior. However, this progressive loading model needs to be optimized since high-load training is required to increase sports performance. For that reason, we propose the use of manual matrix remodeling techniques to optimize the tissue tolerance and increase the fascial system’s capacity for deformation and force absorption before and after return to play.

In regard to this, some specific examples are presented in this subsection.

With regard to return to play in muscle injuries, more specifically calf muscle injuries, it should be remembered that the dynamic adaptation of fibrotic scar tissue and nearby areas may influence muscle tissue lengthening and, consequently, strength performance at longer muscle lengths (Järvinen et al., 2005 B; Opar et al., 2012). For that reason, the authors emphasize the importance of using manual structural techniques applied to the intermuscular connective tissue between soleus and gastrocnemius muscles, as well as between the intermuscular septa of the medial and lateral gastrocnemius muscles to improve the soft tissue’s viscoelastic behavior (Figure 44.3). In this way, the tolerance of the tissue may be increased to load during the lengthening phase of the eccentric contraction, when the susceptibility to sustain muscle distension is higher (Figure 44.4). To show the benefits of these therapeutic techniques, ultrasound elastography has been used to evaluate (in real time) the stiffness and elasticity

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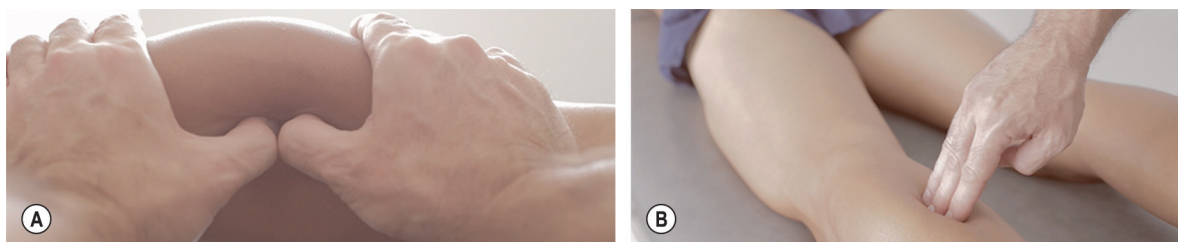


Figure 44.3 Manual matrix remodeling technique

These techniques are based on axial and compressive vectors over the intermuscular septa and/or fibrotic scars to release elastic energy and decrease the excessive stiffness and the pre-tension state of the fascial tissue.

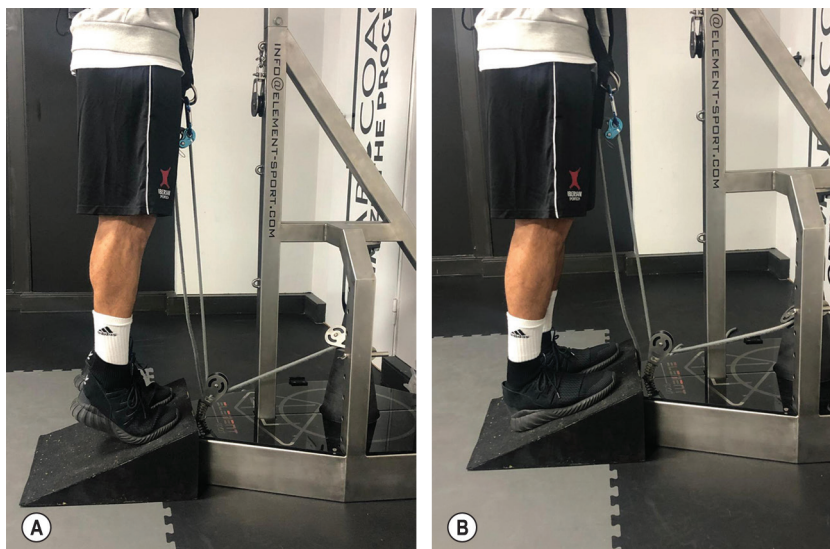


Figure 44.4 High-force eccentric contractions from a lengthened position

Mechanical factors can decrease muscle activity in the lengthened range of eccentric contraction when the susceptibility to sustain a muscle distension is higher. For this reason, we suggest the need to optimize the viscoelastic properties of septa and intermuscular scars using the manual techniques described in the corresponding section before progressive loading exercise programs (i.e. isoinertial training with inclined wedge to increase eccentric muscle strengthening in calf muscles).

of the tissue. Figure 44.5 reveals the elasticity curve registering the myofascial tissue pre-tension during muscle repair and fibrosis (sonoelastographic evolution control). This information is then used to apply the proper treatment, according to the response of the myofascial tissue, after the fascial therapy is performed (Martínez-Rodríguez and Galán del Río, 2013).

In the rehabilitation of chronic tendon injuries, more specifically Achilles tendinopathy, different

authors have studied the multiple benefits derived from the use of eccentric work programs using a progressive overload sequence adapted to the tendon's deformation capacity (Malliaras et al., 2013). However, beyond the protocols of eccentric work and the need to individualize the processes of re-adaptation, it is important to observe the following: in order to increase sports performance, increased loads (stresses) are required. However, at the same time, we also know that

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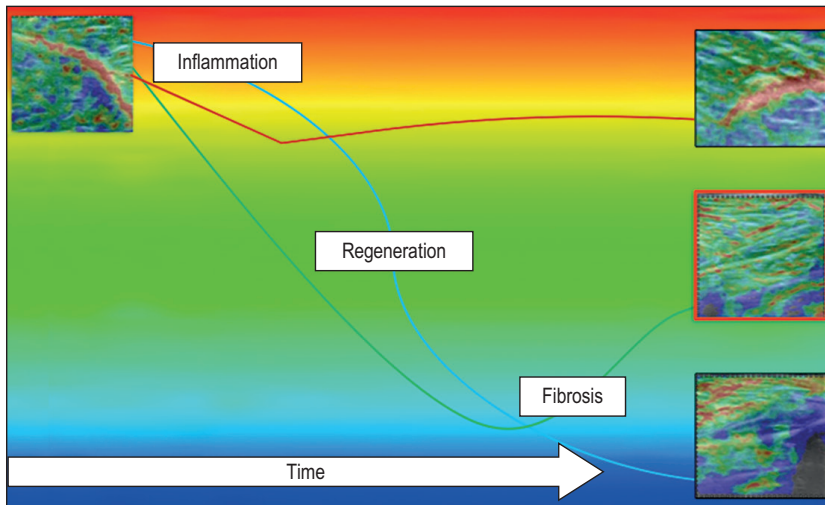


Figure 44.5 Schematic relationship between the physiological scarring process and the elastography chromatic scale

An elasticity curve shows the ideal moment for implementation of manual matrix remodeling techniques to prevent fibrosis. The lines represent the wound-healing process (inflammation, skeletal muscle regeneration and remodeling), which may lead to fibrosis. The green line represents wound healing with restitution of local elasticity after manual modeling of the scar and the blue line shows the wound-healing process with elasticity loss and a high risk of relapse.

From Martínez and Galán del Río (2013) with permission.

the Achilles tendon has a limited deformation capacity (Järvinen et al., 2005 A). In this context, is it possible to increase training and competition loads without conditioning the proper stress-strain relationship? In other words, is it possible to increase the force-absorbing capacity of the Achilles tendon and plantar fascia while taking into account the composition and resistance of these rigid structures (as indicated above)? Understanding the anatomical and functional continuity of the Achilles-calcaneus-plantar system with the triceps surae (Chapter 12), we propose a manual approach of the gastrocnemius-soleus complex (Figure 44.3). Using this approach, it is possible to improve sliding between the fascial layers of the medial head and lateral heads of the gastrocnemius, as well as between gastrocnemius muscles and soleus muscle in the medial and lateral septa

of the lower limb. In other words, by optimizing the mechanical behavior of intermuscular septa, it is possible to improve the adaptability of the Achilles-calcaneus-plantar system and increase the training loads and mechanical efficiency of the tendon.

Finally, core training exercises are usually incorporated in multiple rehabilitation and return to play programs, as core stability is related to an increase of athletic performance and a decrease in re-injury risk (Huxel Bliven and Anderson, 2013). However, a recent study showed that the effectiveness of core training programs, based on the adequate transmission of force between the transverse musculature and the lumbar fascia, may be limited in those cases where there is an increased thickening of the posterior musculofascial junction of the lumbar

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fascia via interfascial lumbar triangle. This interface integrates the aponeurosis of the transversus abdominis with lumbar fascia that bifurcates at this level into the middle and posterior layer of the thoracolumbar fascia (Schuenke et al., 2012; Chen et al., 2016). For that reason, prior to performing core training exercises, we suggest the use of manual techniques in the interfascial lumbar triangle to increase sliding of transverse abdominis during contraction and, subsequently, to increase the functional efficiency of the lumbopelvic region during movements that require eccentric strength and efficient deceleration, as well as fast and explosive changes of direction.

Summary

If fibroblast behavior depends on surrounding mechanical processes, it is relevant to try to fully understand these processes, not only during prevention and treatment of myofascial and tendon

injury, but also in soft tissue, bone and joint injuries after sports-related traumata.

Many of the existing rehabilitation protocols offer programs involving progressive loading exercises. However, based on the authors' clinical experience, this widely used approach would not consider the pre-existing state of collapse and restriction in the fascial network associated with a repetitive overuse pattern, which is common in most sports. Therefore, the authors would like to stress the benefits of initiating any rehabilitation process by using manual therapy on the restricted and fibrotic areas, aimed at encouraging the rearrangement and remodeling of fascial architecture.

In addition, to optimize sports rehabilitation modification of the mechanical properties of the fascial system, alongside mechanical load control, motor strategies should be considered before and after return to play.

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